

A BAYESIAN DENOISING METHOD FOR COMPLEX RADAR SIGNAL WITH APPLICATION TO CLASSIFICATION OF HUMAN INDIVIDUALS

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ABSTRACT

We propose a novel noise-robust classification scheme to discriminate one walking person and two walking persons. Non-parametric Bayesian techniques with the Beta process prior are applied to the Principal Component Analysis (PCA) model for the automatic choice of the number of principal components. Noise reduction is accomplished via reconstructing the echo within the subspace composed of the selected principal components. Then a 2-dimensional feature vector is extracted from the time frequency (TF) spectrograms of the low resolution radar echoes to depict the different micro-Doppler characteristics of one walking person and two walking persons. Compared with the existing CLEAN based noise reduction method, the new method can work without SNR prior information. In the experiments based on measured data, the proposed method shows the good noise reduction and classification performance even under the relative low Signal-Noise-Ratio (SNR) cases.

Index Terms—Micro-Doppler effect, Low resolution radar, Hierarchical Bayesian, Feature extraction, Radar target classification

1. INTRODUCTION

Radar transmits an electromagnetic signal and the signal interacts with a target, based on the time delay between the transmitted signal and the received signal, we can determine the distance between the radar and the target. If the transmitted signal interacts with a moving target, the carrier frequency of the received signal will be shifted from that of the transmitted signal. This phenomenon is known as Doppler Effect. If there are oscillatory or rotation motions in addition to the bulk motion of the target, e.g. the swinging arms of a walking person, the rotating rotor blades of a helicopter, these oscillatory or rotation motions will induce additional sideband frequency components about the

Doppler shift frequency of the returned signal. This additional Doppler modulation is named micro-Doppler effect [1]. Micro-motion and corresponding micro-Doppler effect are highly related to the property of the target, i.e., different kinds of target may have different kinds of micro-motion. Thus the micro-Doppler features can be used to realize the target classification [2, 3].

Since the energy of the micro-motion component in the radar returns from the target is much smaller than that of the translation motion component, it is easily contaminated by the receiver noise component. As shown in the experimental results in [4, 5], classification performance usually descends with the reduction of SNR, and in the low SNR case the performance deteriorates dramatically. Since the SNR directly related to the distance between the target and radar for a given noise power and radar power, noise robustness of a classification method is an important factor in increasing the classification distance between the target and radar in the real application. Reference [6] proposed a simple noise reduction method based on the CLEAN technique and sparse representation theory and it achieves a good noise reduction performance for the measured data from airplane targets. However, this method requires the prior information about the noise power. If the noise power is not estimated accurately, the noise reduction performance will decrease.

This paper reported here proposes an alternative approach based on the principal component analysis (PCA) to accomplish the noise reduction preprocessing procedure in a classification scheme, which is developed via non-parametric Bayesian structure with automatic model selection ability. Then a 2-dimensional time frequency feature vector is extracted from the denoised micro-Doppler signatures to distinguish one or two walking persons. Experimental results based on measured data show the proposed method achieves inspiring denosing and classification performance under relative low SNR.

2. PRINCIPAL COMPONENT ANALYSIS WITH BETA PROCESS

2.1. Bayesian Principal Component Analysis

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Principal component analysis (PCA) [7, 8] is an important data analysis method in statistic pattern recognition which can decompose high-dimensional data into a low-dimensional signal subspace and a noise subspace. In the real application, noise reduction can be accomplished by reconstructing the noised signal within the signal subspace and discarding the residual noise subspace.

Considering a data set of noised observations $\mathbf{X} = \{\mathbf{x}_n\}_{n=1}^N$, with $\mathbf{x}_n \in \mathbb{C}^D$ and $\mathbb{C}^D$ denoting $\mathbf{x}_n$ a complex vector of length D . Noise reduction can be obtained via PCA in the following form:

$$\mathbf{s}_n = \sum_{k=1}^M (\mathbf{x}_n^H \mathbf{u}_k) \mathbf{u}_k \quad (1)$$

where $\mathbf{s}_n$ is the denoised signal and $\mathbf{u}_k, k=1, \dots, M$ denotes the k th eigenvector corresponding to the k th largest eigenvalue of the covariance matrix of the observations. M denotes the dimensionality of the signal subspace.

Therefore, the key problem in PCA for the denosing aim is the choice of the dimensionality of signal subspace, i.e., the value of M . Since M reflects the amount of useful information contained in the observations. If M is chosen smaller than the real one, some useful information will be lost. On the contrary, if it is chosen bigger than the real one, some fake information will be introduced, which can confuse us in some applications.

We propose a non-parametric Bayesian method using a beta process to determine adaptively the dimensionality of the signal subspace. The Bayesian PCA, i.e. BPCA, can be expressed in the following form

$$\mathbf{x}_n = \mathbf{U} \mathbf{Z} \mathbf{U}^H \mathbf{x}_n + \boldsymbol{\varepsilon}_n \quad (2)$$

where $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_D]$ is the $D \times D$ eigenvector matrix while the k th column $\mathbf{u}_k, k=1, \dots, D$ denoting the eigenvector corresponding to the k th largest eigenvalue λ_k . $\mathbf{Z}$ is the diagonal matrix and the k th diagonal element $z_k \sim \text{Bernoulli}(\pi_k)$, $\pi_k \sim \text{Beta}(a_0/K, b_0(K-1)/K)$, where $\text{Bernoulli}(\cdot)$ and $\text{Beta}(\cdot)$ refer to Bernoulli and Beta densities respectively. $\boldsymbol{\varepsilon}_n \sim \text{CN}(0, \gamma^{-1} \mathbf{I}_D)$ is the independent zero-mean complex Gaussian distribution with the covariance matrix $\gamma^{-1} \mathbf{I}_D$. To complete the specification of the hierarchical prior, Gamma prior is placed on the noise precision $p(\gamma) = \text{Gamma}(\gamma | c_0, d_0) = \Gamma(c_0)^{-1} d_0^{c_0} \gamma^{c_0-1} e^{-d_0 \gamma}$, with $\Gamma(c_0) = \int_0^\infty t^{c_0-1} e^{-t} dt$, the 'gamma function'. This Gamma hyper-prior placed on γ is typically non-informative. a_0, b_0, c_0 and d_0 are called 'hyper parameters'.

Since the expectation of π_k is $\langle \pi_k \rangle = \frac{a_0/K}{a_0/K + b_0(K-1)/K}$, by setting K to a large, but finite number, $\langle \pi_k \rangle$ is close to

zero, which means the probability of $z_k = 1$ is very small. According to (2), setting $z_k = 0$ is equivalent to directly removing the k th column, i.e., $\mathbf{u}_k$ from $\mathbf{U}$. In this way, the BPCA model selects the signal subspace adaptively. Base on the above analysis, we now rewrite the BPCA model in the following:

$$\begin{aligned} \mathbf{x}_n &= \mathbf{U} \mathbf{Z} \mathbf{U}^H \mathbf{x}_n + \boldsymbol{\varepsilon}_n, \quad n=1, \dots, N \\ z_k &\sim \text{Bernoulli}(\pi_k), \quad k=1, \dots, D \\ \pi_k &\sim \text{Beta}\left(\frac{a_{0k}}{K}, \frac{b_{0k}(K-1)}{K}\right) \\ \boldsymbol{\varepsilon}_n &\sim \text{CN}(0, \gamma^{-1} \mathbf{I}_D) \\ \gamma &\sim \text{Gamma}(c_0, d_0) \end{aligned} \quad (3)$$

2.2. Variational Bayesian Inference

We derive a variational Bayesian algorithm [9] to perform the fast inference for the proposed BPCA model in (3).

1) The VB-E step

Update for $\mathbf{Z}$

$$p(z_k | \mathbf{x}_n, \mathbf{Z}^{-k}, \gamma^{-1}) \propto p(\mathbf{x}_n | z_k, \mathbf{Z}^{-k}, \gamma^{-1}) p(z_k | \pi_k) \quad (4)$$

where $\mathbf{Z}^{-k}$ is the matrix $\mathbf{Z}$ with the k th diagonal element removed.

The probability that $z_k = 1$ is proportional to

$$\exp(\langle \ln \pi_k \rangle) \times \exp(\langle \gamma \rangle \sum_{n=1}^N \|\mathbf{w}_{nk}\|^2) \quad (5)$$

with $\langle \cdot \rangle$ denoting the expectation and w_{nk} denoting the k th element of $\mathbf{w}_n = \mathbf{U}^H \mathbf{x}_n$. The probability that $z_k = 0$ is proportional to $\exp(\langle \ln(1 - \pi_k) \rangle)$. The expectation mentioned above can be calculated as follows:

$$\langle \ln \pi_k \rangle = \varphi\left(\frac{a_{0k}}{K} + \langle z_k \rangle\right) - \varphi\left(\frac{a_{0k} + b_{0k}(K-1)}{K} + 1\right) \quad (6)$$

$$\langle \ln(1 - \pi_k) \rangle = \varphi\left(\frac{b_{0k}(K-1)}{K} + 1 - \langle z_k \rangle\right) - \varphi\left(\frac{a_{0k} + b_{0k}(K-1)}{K} + 1\right) \quad (7)$$

where $\varphi(\cdot)$ denotes the digamma function. $\langle \gamma \rangle$ is defined in the update for γ .

2) The VB-M step

Update for π

$$p(\pi_k | \mathbf{Z}) \propto p(\mathbf{Z} | \pi_k) p(\pi_k | \frac{a_{0k}}{K}, \frac{b_{0k}(K-1)}{K}) \quad (8)$$

The posterior of π_k can be shown to be

$$\pi_k \sim \text{Beta}\left(\frac{a_{0k}}{K} + \langle z_k \rangle, \frac{b_{0k}(K-1)}{K} + 1 - \langle z_k \rangle\right) \quad (9)$$

Then the expectation of π_k can be obtained

$$\langle \pi_k \rangle = \frac{\frac{a_{0k}}{K} + \langle z_k \rangle}{\frac{a_{0k} + b_{0k}(K-1)}{K} + 1} \quad (10)$$

Update for γ

$$p(\gamma|X, Z) \propto p(X|Z, \gamma^{-1})p(\gamma|c_0, d_0) \quad (11)$$

The posterior of γ can be shown to be

$$\gamma \sim \text{Gamma}(c_0 + DN, d_0 + \sum_{n=1}^N (\mathbf{x}_n^H \mathbf{x}_n - \langle \mathbf{z}^T \rangle (|\mathbf{w}_n|^2))) \quad (12)$$

with $\mathbf{z} = \text{diag}(\mathbf{Z}) = [z_1, \dots, z_D]^T$, $\text{Gamma}(\cdot)$ denoting the Gamma function and

3) The Stopping Criterion

After each iteration, we can decide which column in the eigenvector matrix $\mathbf{U}$ to be selected based on the value of $\mathbf{z} = \text{diag}(\mathbf{Z}) = [z_1, \dots, z_D]^T$, i.e., if $z_k > 0.5, k = 1, \dots, D$ the corresponding k th column in $\mathbf{U}$ is selected. Then after the i th iteration, we can reconstruct the denoised signal using (1), i.e., $\mathbf{s}_n^i = \sum_{k=1}^q (\mathbf{x}_n^H \mathbf{u}_k) \mathbf{u}_k$ where q equals the number of columns selected in $\mathbf{U}$ after the i th iteration. If the error between the reconstructed signal after the i th iteration, i.e., $\mathbf{s}_n^i$ and the $(i+1)$ th iteration, i.e., $\mathbf{s}_n^{i+1}$ falls below a small threshold (e.g., 10^{-8}), the algorithm stops. Then the denoised signal can be reconstructed using the subspace composed of the selected principal components

$$\mathbf{s}_n = \sum_{k=1}^M (\mathbf{x}_n^H \mathbf{u}_k) \mathbf{u}_k \quad (13)$$

3. TIME-FREQUENCY FEATURE EXTRACTION

Since our measured data are collected under cooperative measurement experiments with a low resolution radar, which works at a Ka waveband, the high SNR condition can be guaranteed and $\text{SNR} \approx 35\text{dB}$. Measured data are collected under two experimental scenarios: 1) a single person walking towards or away from radar; 2) two persons walking towards or away from the radar and their walking speed is close but not the same. The dwell time is 1.2s and the pulse repetition frequency is 3 KHz. The short time Fourier transformation is then applied to the data to generate the corresponding spectrogram. Fig. 1(a) and (b) show the TF spectrogram of the measured data from one person and two persons.

Here we extract the following two features from the TF spectrogram to depict the different micro-Doppler signatures between the radar returns from one person or two persons in order to accomplish the classification task.

The first feature: the energy entropy of the average Fourier spectrum

$$f1 = -\sum_{f=1}^F \rho_f \ln \rho_f \quad (14)$$

with

$$\rho_f = |G(f)|^2 / \sum_{f=1}^F |G(f)|^2 \quad \text{and}$$

$$G(f) = \text{FFT}(g(t)) = \text{FFT}\left(\frac{\sum_{f=1}^F f \cdot |S(t, f)|}{\sum_{f=1}^F |S(t, f)|}\right), \quad \text{which}$$

represent the energy entropy of the average Fourier spectrum $G(f)$. In the TF spectrogram $S(t, f)$, each column denotes the instantaneous frequencies of all the scatters on the target. Thus $g(t) = \frac{\sum_{f=1}^F f \cdot |S(t, f)|}{\sum_{f=1}^F |S(t, f)|}$ can

represent the average instantaneous frequency component of the target signal at time t . Therefore, we can obtain all the frequency components in the time domain $[1 \ T]$ via the Fourier transform, i.e., the average Fourier spectrum $G(f) = \text{FFT}(g(t))$ with $\text{FFT}(\cdot)$ denoting the fast Fourier transform.

The second feature: the maximum value of the normalized value of average Fourier spectrum

$$f2 = \max(\rho) \quad (15)$$

with $\rho = \{\rho_f\}_{f=1}^F$. This feature denotes the energy distribution of ρ , the more the energy of ρ gathers, the larger the maximum value is.

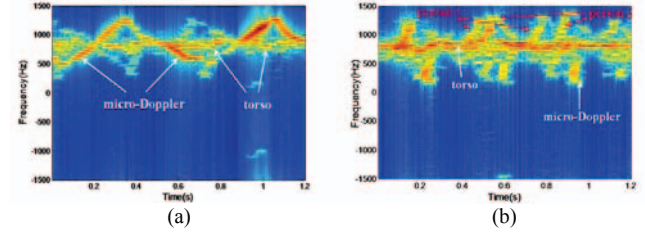


Fig. 1. Time-frequency spectrograms of measured echoes from walking human individuals under the SNR condition about 35dB: (a) one person; (b) two persons.

4. NOISE REDUCTION AND CLASSIFICATION PERFORMANCE ON MEASURED DATA

Fig.2 shows the noise reduction results for the measured data from one or two walking persons of the proposed BPCA method, the CLEAN based method with accuracy SNR information and the CLEAN based method with the 3dB biased SNR information. The TF spectrograms of one walking person and two walking persons under the SNR condition of 10dB are shown in Fig.2 (a) and (b) respectively. Fig.2 (c) and (d) depict the TF spectrum of the denoised signals from Fig.2 (a) and (b) via the proposed BPCA method. Fig.2 (e) and (f) respectively show the TF spectrogram of the denoised signals from Fig.2 (a) and (b) via the CLEAN based method with the accuracy SNR information [6]. Obviously, the proposed BPCA method can not only reduce the noise component well but also preserve the original micro-Doppler signal well, while there are noise

components left in the TF spectrograms of the signals denoised via the CLEAN based method.

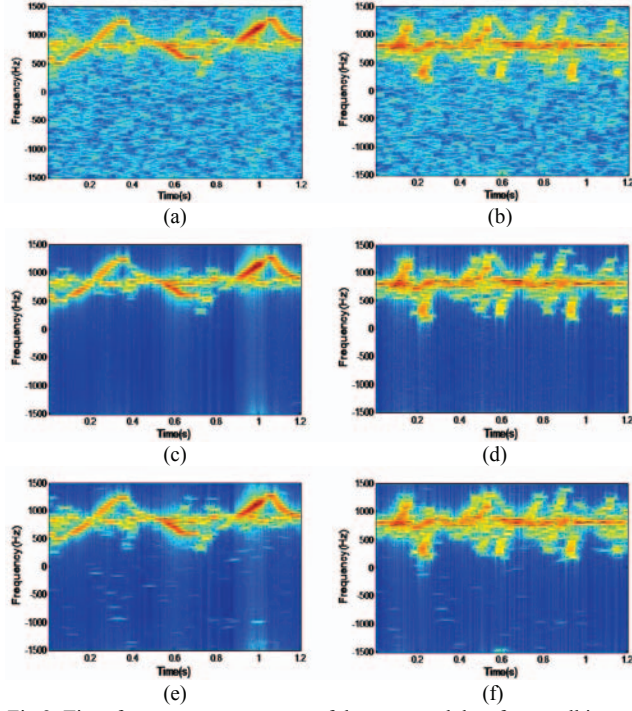


Fig.2. Time-frequency spectrogram of the measured data from walking persons: (a) one person, SNR = 10dB; (b) two persons, SNR = 10dB; (c) denoised results from (a) via BPCA; (d) denoised results from (b) via BPCA; (e) denoised results from (a) via CLEAN based method with accuracy SNR information; (f) denoised results from (b) via CLEAN based method with accuracy SNR information.

In the classification experiment, we utilize the measured data described in Section 3 to evaluate the classification performance of the proposed noise-robust classification scheme. Since our measured data are collected under high SNR condition ($\text{SNR} \approx 35\text{dB}$), we use half of them in the training stage to extract the 2-dimensional feature vector $F = [f_1, f_2]$ which is then used to train the classifier. In order to evaluate the noise reduction performance of the proposed BPCA method, we add simulated complex zero-mean Gaussian noise corresponding to the different SNR cases to the left half of measured data, which are used as the test data. In the test stage, we use the proposed BPCA based noise reduction method to denoise each noised test sample, then the feature vector $F = [f_1, f_2]$ is extracted for each denoised test sample. Finally, the classifier learned in the training stage is used to give a classification decision for each test feature vector. In our experiment, we choose the support vector machine (SVM) with Gaussian kernel as the classifier. The parameter of Gaussian kernel is tuned in the range $[0, 5]$, and we select the best classification result from those of the different parameter settings for the following analysis and comparison.

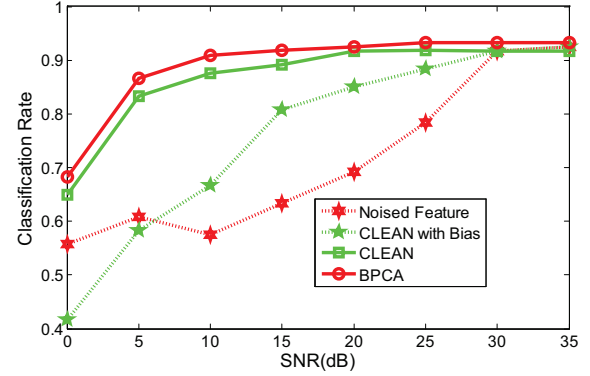


Fig.3. Variation of the classification accuracy rates of the measured data with the test SNR via noised features, CLEAN with accuracy SNR information, CLEAN with 3dB biased SNR information and BPCA. Half of the measured data, which are collected under high SNR condition ($\text{SNR} \approx 35\text{dB}$), are used as the training data, and the left half, which are added simulated noise, are used as the test data.

Fig.3 depicts the classification accuracies of different methods versus the test SNR. In Fig.3 the "Noised Feature" denotes the classification accuracies of the features described in Section 3 and extracted from the noised test data without noise reduction preprocessing; the "CLEAN" denotes the classification accuracies of the features denoised via the CLEAN based noise reduction method with the accuracy SNR information[6]; the "CLEAN with bias" denotes the classification accuracies of the features denoised via the CLEAN based noise reduction method with the 3dB biased SNR information[6] and the "BPCA" denotes the classification accuracies of the features denoised via the BPCA based noise reduction method. From Fig.3 we can see that our new robust classification scheme always performs best in distinguishing one walking person and two walking persons under the different SNR conditions. If we assume that the classification threshold for the real application is 90%, our proposed noise robust method can work under the test condition of $\text{SNR} > 10\text{dB}$; the CLEAN based methods required $\text{SNR} > 17\text{dB}$ and $\text{SNR} > 20\text{dB}$, respectively.

5. CONCLUSION

In this paper, we propose a novel noise-robust classification scheme to discriminate of one walking person and two walking persons via the 2-dimensional TF feature vector extracted from the micro-Doppler signatures of their low resolution radar returns. In order to improve the classification performance under the low test SNR conditions, we propose a nonparametric extension to the principal component analysis problem via a Beta process prior for the test sample denosing. We demonstrate our noise robust classification scheme on measured data and experiment results show that the proposed method can achieve good noise reduction and classification performance under the relatively low test SNR conditions.

6. REFERENCES

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